MIL-W-52500C
1 August 1978
SUPERSEDING
MIL-W-52500B
21 September 1973

MILITARY SPECIFICATION

WATER PURIFICATION UNIT, ELECTRIC-MOTOR-DRIVEN, ERDLATOR-TYPE CLARIFIER, DIATOMITE-TYPE FILTER, VAN-BODY-MOUNTED, 3,000 GPH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a military design electric-motor-driven, van-body mounted, 3,000 gpm water purification unit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-C-596/57	- Connector, Plug, Electrical, Specific Purpose, Locking, 4 Pole, 4 Wire, 20 Amperes, 250 Volts.

FSC 4610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-W-6858	- Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-H-11511	- Heaters, Space: Multifuel, with Blower, 60,000 BTU/HR.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52286/25	- Cable Assembly, Power Electrical, Adapter.
MIL-C-52286/27	- Cable Assembly, Power Electrical, 4 Conductor, One End W/Watertight Male Plug Connector, Other End W/Watertight Female Plug Connector and Watertight Male Dummy Plug Connector.
MIL-S-52333	- Sleeves, Filter-Element, Water Purification Units.
MIL-W-52619	- Water Purification Units, and Water Purification Equipment Sets, 3,000 GPH; Packaging of.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS16203	- Nut, Plain, Hexagon, and Hexagon Jam, UNC-2B, Nonmagnetic.
MS35307	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt) Steel, Corrosion-Resisting, Passivated, UNC-2A.
MS35842	- Clamp, Hose - Low Pressure Type "F".
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS51393	- Chassis, Truck (Military Design) M46C, 2-1/2 Ton, 6X6, 190 Inch Wheelbase.

DRAWINGS

ME

TA13208E9599	- Water Purification Unit, Electric-Motor-Driven, Erdlator-Type Clarifier, Diatomite-Type Filter, Van-Body Mounted, 3,000 GPH.
D13208E5820	- Terminal Load.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN WELDING SOCIETY

D1.1 - Structural Welding Code, Section 5, Qualification.
A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

ANS B16.3 - Malleable-Iron Screwed Fittings, 150 and 300 Lb.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A123 - Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars, and Strip.
A575 - Merchant Quality Hot-Rolled Carbon Steel Bars.
A576 - Special Quality Hot-Rolled Carbon Steel Bars.
B173 - Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors.
B209 - Aluminum-Alloy Sheet and Plate.
B211 - Aluminum-Alloy Bars, Rods, and Wire.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The water purification unit shall be in accordance with Top Assembly TA13208E9599 and as specified herein. Unless otherwise specified (see 6.2), the contractor shall mount the unit on a Government-furnished, 2-1/2 ton, M46 truck chassis (see 3.16).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more water purification units, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce water purification units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more water purification units for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings (see 6.12). Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The material shall have no adverse effect on the health of personnel when

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used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting officer to the appropriate department medical service who will act as an advisor to the contracting officer (see 6.4).

3.5 Operation. The water purification unit shall operate without any leaks, malfunction of any component, or any permanent deformation.

3.6 Van body. The van body, including doors and side panels, shall be watertight and shall prevent internal lights from being seen under blackout conditions. The doors and side panels shall open, close, and latch without binding. The side panels shall require not more than two persons to open, close, and latch. When switches are set for blackout operations, lights inside the van body shall go off before doors clear the jamb.

3.7 Transportability. The water purification unit shall be capable of withstanding the shock and vibrational stress encountered during transportation.

3.8 Environmental.

3.8.1 Operating temperature. The water purification unit shall perform as specified in ambient temperatures from plus 110° F to minus 25° F.

3.8.2 Storage temperature. The water purification unit shall not be damaged by storage at ambient temperatures from plus 160° F to minus 30° F.

3.9 Reliability. The specified Mean-Time-Between-Failure (MTBF) of the water purification unit and ancillary equipment shall be 200 hours when tested as specified in 4.6.2.8.

3.10 Maintainability. Each maintenance assembling and disassembling operation performed as a result of testing in accordance with 4.6.2.9 shall be accomplished by not more than two men using common tools furnished with the water purification unit. The ratio of manhours of maintenance required to the hours of operation shall not exceed 0.08. A maintenance schedule shall be furnished prior to start of any testing.

3.11 Electromagnetic interference. The electromagnetic interference emission characteristics of the water purification unit shall conform to MIL-STD-461, Table A-I, Tactical Support Class for Test Method RE02 and A-CE2.

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3.12 Accessories. The following accessories shall be furnished with each water purification unit in the quantities shown:

Item	Quantity
Cable assembly conforming to Figure 1.	2
Cable assembly conforming to Specification Sheet MIL-C-52286/27. Cable D of MIL-C-52286/27 shall be CO-04 HGF (4/12) 0665 or UL Type SO Cable, Neoprene Jacket, 600V, 60° C, Type SO-4, (4/12) (65 x 30)	3
Paddle conforming to Figure 2	1
Blind flange conforming to Figure 3.	6
Pipe cap conforming to ANS B16.3, 1/2-inch nominal pipe size.	6
Hexagon head capscrew conforming to MS35307-362.	12
Hose clamp conforming to MS35842-15.	12
Float conforming to Figure 4.	1
Ground rod conforming to Figure 5.	2

3.13 Lubrication. All surfaces not in contact with the water being treated requiring lubrication shall be provided with a means for lubricating.

3.13.1 Lubricants. The water purification unit shall operate as specified herein when lubricated with military lubricants (see 6.6).

3.13.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.13.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.13.4 Filling, draining, and checking provisions. Enclosures such as gear cases that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drain-plug conforming to MS35844 or MS49006. The drain outlet shall be located

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so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs shall be used to convey the lubricant from the drain to a point where it can be collected in a container. The drainplug, the filling means, and the lubricant level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.13.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, electronic gear, and mechanical control systems; for use on gears, actuator screws, and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

(b) MIL-G-10924 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of minus 65° F to plus 225° F. Not for use on machinery which comes in contact with potable water.

The water purification units shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.13.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), and upon completion of all testing, all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to plus 32° F. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.14.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.14.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102 - Southern California Air Pollution Control District.

3.15 Identification marking. The water purification unit shall be identified in accordance with MIL-STD-130 and as specified on applicable drawings.

3.16 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.7):

Item No.	Description	Identification	Quantity for each
1	Chassis, truck; 2-1/2 ton 6 x 6, w/winch, M46.	MS51393-1	1 (Unless otherwise specified) (see 3.1)
2	Sleeves, filter element.	MIL-S-52333, size 2.	12
3	Heater, space, multi-fuel with blower, 60,000 Btu.	MIL-H-11511, Type I, air discharge to side.	1

3.17 Noise level. The noise level of the water purification unit (excluding horns and warning alarms) shall be in accordance with MIL-STD-1474 Category D for steady-state noise when tested in accordance with 4.6.2.11.

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3.18 Safety. A grounding system shall be incorporated to insure safety from static electricity.

3.19 Human factors. The characteristics of the water purification unit shall provide for operation by personnel in all type clothing, and shall conform to MIL-STD-1472.

3.20 Workmanship. All parts, components, and assemblies of the water purification unit including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered. The unit shall be free from such defects as misalined components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and serviceability of the unit.

3.20.1 Metal fabrication. Metal used in the fabrication of the water purification units shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Sheared, sawed, and clipped edges shall be free of burrs and slivers. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing for steel only. Methods utilizing heat shall not be used for nonferrous metals. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Holes shall be punched or drilled and shall have the burrs removed.

3.20.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.20.3 Riveted connections. Rivets shall fill the holes completely. All rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.20.4 Welders, weldments, and welding.

3.20.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests

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as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Welders who only make horizontal welds need not qualify for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds.

3.20.4.2 Weld filler metals. Aluminum filler metal shall be in accordance with AWS A5.10, Type ER5356 or ER5556, except that filler metal ER4043 may be used in highly restricted locations if conditions cannot be eliminated by adjustments of procedures or fixtures, or when joining base metal Type 6061 to other types of aluminum. Steel filler metal shall be in accordance with AWS A5.1, Type E6010 or E6011 for carbon steel; Type E7018 for high-strength, low-alloy steel; Type E6018 for high-strength, low-alloy steel to carbon steel; and AWS A5.9, Type ER308L for corrosion-resisting steel.

3.20.4.3 Aluminum and steel weldments. The surfaces and edges of parts to be welded including layers of weld materials when multiple layers are required shall be free from oxide scale, paint, lubricant, and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities and other imperfections that may reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified on the applicable drawings.

3.20.4.3.1 Aluminum welding. Aluminum welding, except spot and seam welding, shall be done by the metal-inert-gas (MIG) method or other methods that will provide equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.20.4.2. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic or nonmetallic inclusions. Finished aluminum welds may be ground, filed, chipped, or wire-brushed, but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 300° F for a total time not more than 30 minutes. No process requiring the use of welding flux shall be used. Work shall be positioned and jigged for flat welding whenever practicable.

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3.20.4.3.2 Steel welding. Steel welding shall be done by the shielded-electric-arc method or metal-inert-gas (MIG) process in accordance with qualification procedures described in 3.20.4.1, or by any other method providing equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.20.4.2. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. Work shall be positioned for flat welding whenever practicable. Complete and uniform pentration and fusion of the metals shall be obtained on all welds.

3.20.4.4 Weldment specimens. The contractor shall submit steel and aluminum specimens representing each typical weld joint utilizing in fabrication of the water purification unit. These specimens shall be furnished at the same time as the preproduction model water purification unit. Weldment specimens shall be submitted by each welder performing weldments on the water purification unit with each welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as an inspection standard for evaluation of production weldments.

3.20.4.5 Welding procedure. Prior to any welding, the contractor shall prepare and submit to the contracting officer for approval a written welding procedure defining process and control. This procedure shall include, but not be limited to, the factors listed in Table I. Welding procedure changes affecting any of the factors listed in Table I shall be submitted to the contracting officer for approval prior to adoption of the change.

Table I. Factors to be Included in Joint Welding Procedure

Item No.	Factor
1.	Welding process and techniques including shielding medium.
2.	Welding process operation.
3.	Type of current, ac or dc; and polarity of dc, if used.
4.	Shielding gas composition.
5.	Joint design.
6.	Method of joint preparation.
7.	Preweld and interpass cleaning.
8.	Preheat or ambient temperature range.
9.	Composition of filler metal.
10.	Filler metal size for all passes.
11.	Number and sequence of passes.
12.	Welding position.
13.	Repair method to be used in removing defects.
14.	Method of inspection used to insure proper repair.
15	Method of qualification of welders.
16.	Method of determining capability of automatic equipment and equipment operators.
17.	Workmanship specimens.
18.	Method of inspection of finished welds.

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3.20.4.6 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects specified in 3.20.4.3 or not conforming to the weldment specimens specified in 3.20.4.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Repairs in the form of patching aluminum shall not be performed.

3.20.4.7 Jigs and fixtures. Components except minor parts shall be assembled in jigs or frames and welded while held in proper position. Jigs or frames shall be designed to minimize distortion of the parts being welded.

3.20.4.8 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858, Class C.

3.20.5 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall be applied to obtain a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

3.20.6 Cleaning. The water purification unit shall be thoroughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the water purification unit unsuitable for its intended use. The filter element shall be disassembled for cleaning, dried, and reassembled for packaging. The unit shall be completely drained before packaging.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Water purification units which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

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4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.9 and 6.10.1).

4.1.5 End Item Final Inspection Requirement (EIFIR). Each water purification unit shall be inspected in accordance with EIFIR-EF-13208E9599. One copy of the complete EIFIR shall be shipped with each inspected water purification unit (see 6.10.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction model inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction water purification unit shall be examined for the defects marked "X" in Column 2 of Table II. Presence of one or more defects shall be cause for rejection or shall be cause for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the preproduction model shall be subjected to the tests marked "X" in Column 2 of Table III. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.3.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production water purification units will be selected at random by the Government from the water purification units being produced by production tooling and will be examined for the defects marked "X" in Column 3 of Table II and subjected to the tests marked "X" in Column 3 of Table III, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production water purification unit shall not exclude the remaining water purification units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production water purification unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production water purification unit and shall be cause for refusal by the Government to continue acceptance of production water purification units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all water purification units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level III.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for defects marked "X" in Column 4 of Table II. AQL shall be 1.5 percent defective.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table III. AQL shall be 0.25 percent defective.

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4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table II.

Table II. Examination Schedule

Ind. Comp.	Pre-prod.	Init. Prod.	Qual. Conf.	Defect	Requirement
1	2	3	4	5	6
X	-	-	-	101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.	Applicable dwgs.
X	-	-	-	102. Any dimensions, other than those identified by the SQAP requirements shown on the drawings, not as specified.	Applicable dwgs.
-	X	X	X	103. Assembly not complete or not correct.	Applicable dwgs.
-	X	X	X	104. Parts or components missing.	Applicable dwgs.
-	X	X	X	105. Parts or components not as specified.	Applicable dwgs.
-	X	X	X	106. Material not as specified.	Applicable dwgs. Figs 1 thru 5
-	X	X	X	107. Treatment and painting not as specified.	Applicable dwgs.
-	X	X	X	108. Lubrication devices not as specified.	3.13
-	X	X	X	109. Workmanship not as specified.	3.20
-	X	-	X	110. Water purification unit not completely drained and cleaned before packaging.	3.20.6
-	X	X	X	111. Accessories missing.	3.12
-	-	X	X	112. Electromagnetic interference suppression components and network not identical with approved model.	3.11

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4.6.2 Tests.

4.6.2.1 Test schedule. Tests shall be in accordance with Table III.

Table III. Test Schedule

Ind. Comp. 1	Pre-prod. 2	Init. Prod. 3	Qual. Conf. 4	Test 5	Requirement 6	Test Paragraph 7
X	-	-	-	Various.	See applicable drawings.	4.1.2
-	X	-	-	Operational (6 hours)	3.5	4.6.2.3
-	-	-	X	Operational (2 hours)	3.5	4.6.2.4
-	X	-	X	Van body.	3.6	4.6.2.5
-	-	X	-	Transportability.	3.7	4.6.2.6
-	-	X	-	Environmental.	3.8	4.6.2.7
-	-	X	-	Reliability (MTBF).	3.9	4.6.2.8
-	-	X	-	Maintainability.	3.10	4.6.2.9
-	X	-	-	Electromagnetic interference.	3.11	4.6.2.10
-	-	X	-	Noise level.	3.17	4.6.2.11
-	-	X	-	Safety.	3.18	4.6.2.12
-	-	X	-	Human factors.	3.19	4.6.2.13

4.6.2.2 Test conditions. Prior to test, the water purification unit shall be lubricated with oils and greases specified in 3.13. Oils shall be those designated for use in the ambient temperature at the place of test.

4.6.2.3 Operational test (6 hours).

4.6.2.3.1 Test equipment. Test equipment shall include but shall not be limited to the following:

Power source	- 120/208-volt, 3-phase, 4-wire, 60-Hz.
Centrifugal pumps	- 65-gpm at 50-foot head, 2-hp, electric-motor-driven; 208-volt, 60-Hz., ac, 1-1/2-inch suction and discharge.
Suction hose	- NPSH couplings.
Ferric chloride	- Anhydrous, commercial.
Limestone	- Pulverized, commercial.

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Calcium hypochlorite	- Commercial, 65 percent available chloride by weight, minimum.
Filter aid	- Johns-Manville 535, Great Lakes Carbon Dicalite 4200, or equal.
Water	- Tap (potable).

4.6.2.3.2 Preparation of unit. Connect the unit to the power source. Connect a centrifugal pump between the water source and the raw-water inlet with suction hose. Charge and set the chemical feeders to deliver the ppm as specified in Table IV. Set the feed rate in accordance with the calibration charts.

Table IV. Charge and Feed Rates of Chemical Feeders

Feeders	Charge	Feed rate (ppm)
Ferric chloride	5 gallons of solution	20
Calcium hypochlorite	5 gallons of solution	5
Diatomaceous silica	No charge (see 4.6.2.3.4, items (r) and (s)).	0
Limestone	14 gallons of slurry	50

4.6.2.3.3 Procedure for starting water-treating section. Start the water-treating section using the following procedures:

(a) Prime raw-water pump.
(b) Open raw-water aspirator valves.
(c) Open drain valve on clear well.
(d) Close bypass valve on raw-water influent line, erdlator drain valves, and valves on suction lines of filter.
(e) Turn on raw-water pump.
(f) As soon as raw water enters the erdlator, start feeders and agitators, and adjust rate to 50 gpm as read on flow controller.
(g) Adjust valve on raw-water line to limestone feeder aspirator to insure takeup and flow of limestone to erdlator.

4.6.2.3.4 Procedure for starting and backwashing filter. Start and backwash each filter using the following procedure:

Starting

(a) Close drain on clear well of water-treating section.
(b) Close filter and filter-pump drain valves.

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(c) Turn air relief valve and influent plug valve to filter operation.
(d) Turn effluent plug valves to recirculation operation.
(e) Close drain valves on pressure gages.
(f) Close gate valve located below flow controller.
(g) Open filter vent valve located above precoat tank.
(h) Pour 1/2 gallon of water into precoat tank.
(i) Close precoat tank valve after water has drained from tank.
(j) Open valve on influent lines of filter.
(k) Turn on buzzer alarm switch.
(l) Turn on filter pump.
(m) When water starts escaping through filter vent valve, close valve.
(n) When the effluent pressure gage reads 30 psi, open gate valve located below flow controller.
(o) Allow filter to operate on recirculation for not less than 5 minutes or more than 10 minutes.
(p) Set flow controller for 25 gpm.
(q) Turn effluent plug valve for filter operation.
(r) Open valve on filter affluent line to diatomaceous silica feeder, fill feeder to high-level mark, and charge with 7 pounds of diatomaceous silica. Use filtered or settled water.
(s) Set feed rate of diatomaceous silica feeder for 30 ppm in accordance with calibration charts.

Backwashing

(a) Close gate valve located below flow controller.
(b) Operate filter pump for 15 to 30 seconds, then shut off filter pump.
(c) Close valve on suction line to pump.
(d) Turn air relief valve to backwash and then immediately open filter drain valve.
(e) After filter has drained, turn influent plug valve to wash position, open suction line to pump, turn on pump, and wash filter for 1 to 2 minutes.

4.6.2.3.5 Test procedure. Start the water-treating section as specified in 4.6.2.3.3. After the water-treating section is filled with water, start the filter (see 4.6.2.3.4), and operate the complete unit until not less than twelve 1/2-hour starting-and-backwashing cycles on each filter have been accomplished. A 1/2-hour starting-and-backwashing cycle shall start upon completion of item (a) under "Starting" of 4.6.2.3.4 and end upon completion of item (e) under "Backwashing" of 4.6.2.3.4. Short shutdown periods

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for inspection, lubrication, or functional changes will be permitted provided normal operation is resumed within 5 minutes. Upon completion of the last backwashing, shut down and examine the unit. Remove the pumping heads of the solution feeder and examine for damage. Remove the filter dome and examine the elements for damage.

4.6.2.3.6 Criteria for failure. Evidence of one or more of the following conditions shall constitute failure of this test:

(a) Leaks at any welded seam or any connection.
(b) Light, buzzer, or any switch fails to operate.
(c) Low-level alarm fails to operate within the water-level limits.
(d) Valves fail to operate properly.
(e) Evidence of water or air escaping through air-release valve, or vent valve.
(f) Pumping heads of solution feeder corroded.
(g) Mechanical or electrical failure of any component or part.

4.6.2.4 Operational test (2 hours). The water purification unit shall be tested as specified in 4.6.2.2 through 4.6.2.3.6 except as follows:

(a) Use tap water in lieu of ferric chloride solution, calcium hypochlorite solution, diatomite slurry, and limestone slurry.
(b) Operate the complete unit for only four 1/2-hour starting-and-backwashing cycles in lieu of 12 cycles.
(c) Item (f) of criteria for failure is not applicable.

4.6.2.5 Van body tests. The following tests shall be conducted, and may be conducted with the water purification unit removed from the truck chassis. The truck chassis may be leveled prior to conducting the van body tests by installing blocks or timbers under the wheels.

4.6.2.5.1 Level test. Level the water purification unit until the bubble in each of the three levels on the outside of the van body is between the two marks. A difference in height of more than 1/4 inch in the transverse direction or of more than 1/2 inch in the longitudinal direction shall constitute failure of this test.

4.6.2.5.2 Door tests. Level the water purification unit as specified in 4.6.2.5.1. Then fully open and close all doors and side panels for not less than five cycles of opening and closing. Use not more than two persons when opening and closing the side panels. Inability to open, close, and latch the doors and side panels without binding shall constitute failure of this test.

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4.6.2.5.3 Blackout test. Perform this test at night or in an area devoid of daylight or artificial light. Level the water purification unit as specified in 4.6.2.5.1. Close all doors and side panels of the van body, switch on interior lights for normal operation, cover the windows with the shields, and tie shut the cable outlet with the blackout covers. Examine for light leaks at a distance of from 5 to 6 feet from the van body. After examination, set the light switches on blackout and unlatch the doors. Perform the test for light switches actuated by the doors using both the 24-volt and 120-volt circuits. Evidence of any light seen from outside the van body or failure of the lights to go out before the door clears the jamb shall constitute failure of this test.

4.6.2.5.4 Watertightness test. Level the water purification unit as specified in 4.6.2.5.1. Apply a simulated rainfall having a rate of not less than 3 inches per hour and falling at an angle of 45 degrees plus or minus 5 degrees from the vertical simultaneously to the roofs, sides, and ends of the van body for not less than 1 hour. Evidence of any water leakage into the van body shall constitute failure of this test.

4.6.2.6 Transportability. Water purification units shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure VI, and Method 514.2, Procedure XIII, Part 2, except the item shall be impacted once in each direction at a speed of 8 miles per hour $\pm$ 5 percent (total of two impacts). Tests shall be conducted with the unit mounted on a M46 truck chassis. Any shifting of contents, any loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the water purification unit shall constitute failure of this test.

4.6.2.7 Environmental.

4.6.2.7.1 High temperature. The water purification units shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The maximum test temperature for storage shall be 160° F and for operating shall be 125° F. Filter element sleeves shall be installed in the filter in accordance with the applicable technical manual furnished with the unit. Operation shall be in accordance with the 2-hour operational test specified in 4.6.2.4. One or more of the following shall constitute failure of this test:

(a) Leaks at any welded seam or connection.
(b) Lights, buzzer, or any switch fails to operate.
(c) Valves fail to operate properly.
(d) Evidence of water or air escaping through air-release valve or vent valve.
(e) Mechanical or electrical failure of any component or part.
(f) Cracks or splits in any filter element sleeve.

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4.6.2.7.2 Low temperature. The water purification units shall be tested in accordance with MIL-STD-810, Method 502, Procedure I. The minimum test temperatures for storage shall be minus 30° F, and the minimum operating temperatures shall be minus 25° F. The personnel heater in the van body shall be set to operate when the air temperature in the van body falls below plus 32° F. Operation shall be in accordance with the 2-hour operational test specified in 4.6.2.4. One or more of the following defects shall constitute failure of this test:

(a) Leaks at any welded seam or any connection.
(b) Lights, buzzer, or any switch fails to operate.
(c) Valves fail to operate properly.
(d) Evidence of water or air escaping through air-release valve or vent valve.
(e) Mechanical or electrical failure of any component or part.
(f) Cracks or splits in any filter element sleeve.
(g) Failure of heater to maintain air temperature inside the van body at plus 32° F or more.

CAUTION. Prior to start of test and immediately after 2-hour operation, all valves shall be opened and the unit drained of water.

4.6.2.8 Reliability. Using the MTBF specified in 3.9, the water purification unit shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan II. Continue testing until either an "accept" or "reject" decision is reached. Failure, for the purpose of this test, is defined as any malfunction which the operators cannot remedy by adjustment, repair, or replacement action within 30 minutes (excluding waiting or start-up time), utilizing the resources available to the parent unit or organization when deployed, and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance capabilities of the system.
(b) Serious damage to the system by continued operation.
(c) Serious personnel safety hazard.

Simultaneous related malfunctions are considered as one failure.

4.6.2.8.1 Test procedure. The water purification unit with ancillary equipment shall be set up, operated, and maintained in accordance with the applicable technical manual supplied with the unit. The ancillary equipment shall consist of all items listed in the latest issue of SC4610-97-CL-E06. Commence operation using the procedures outlined in 4.6.2.3.3 and 4.6.2.3.4. Refer to the technical manual for complete operational instructions in order to simulate actual use. Operate equipment continuously for

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periods of 20 hours per day at rated capacity, then shut down unit. Daily preventive maintenance and cleaning is allowed during the 4-hour non-operating period (not to be included in assessing MTBF).

4.6.2.8.2 Failure criteria. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.9 Maintainability. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.10. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.10 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.11. The contractor shall furnish the contracting officer a report of tests, from an independent tester, required by MIL-STD-461 (see 6.5). Disapproval of the report shall constitute failure of this test.

4.6.2.11 Noise level.

4.6.2.11.1 Operators station. Measure the sound level at the operators station (inside van body) in accordance with MIL-STD-1474 for steady-state noise while the unit is operating at rated capacity. Nonconformance to 3.17 shall constitute failure of this test.

4.6.2.11.2 Exterior spectator position (outside van body). Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.17 shall constitute failure of this test.

4.6.2.12 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.18 shall constitute failure of this test.

4.6.2.13 Human factors. The water purification units shall be evaluated to determine the human factor characteristics. Nonconformance to 3.19 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select water purification units at any time during the contract production period and subject these water purification units to the examinations marked "X" in Column 3 of Table II and the tests marked "X" in Column 3 of Table III, to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production water purification units. In addition to any test specified as part of the inspection comparison, the Government reserves the right

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to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison water purification unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison water purification unit and shall be cause for refusal by the Government to continue acceptance of production water purification unit(s) until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all water purification units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-W-52619.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and mark- shall be in accordance with MIL-W-52619. The degree of preservation shall be Level A, Level B, or Commercial, and the degree of packing shall be Level A or Commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The water purification unit is intended to supply 3,000 gph of potable water from a raw, fresh-water source.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When the water purification unit shall not be truck chassis mounted by the contractor (see 3.1).
(c) Number of preproduction models required, and time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests,

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the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

(e) When initial production inspection is required, and number of initial production water purification units to be furnished, when applicable (see 3.3).

(f) Degree of preservation and degree of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production water purification units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish units conforming to this specification.

6.4 Medical advisor. For Army procurement, the Surgeon General of the Army will act as advisor to make the necessary determination on health hazards (see 3.4).

6.5 Electromagnetic interference. The contracting officer should forward a copy of the test report (see 4.6.2.10), to the Commander, US Army Mobility Equipment Research and Development Command, ATTN: DRDME-GE, Fort Belvoir, VA 22060.

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the water purification unit covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.7 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.16.

6.8 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examination and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

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6.9 Definitions.

6.9.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.9.2 End Item Final Inspection Requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance, and shall reflect all approved Engineering Change Proposal (ECP), Request for Waivers (RFW), and Request for Deviations (RFD).

6.10 Data requirements. The contracting officer should include requirements for such data as technical publications, instructural materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each water purification unit.

6.11 Provisioning. The contracting officer should include provisioning requirements for such repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of water purification units.

6.12 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

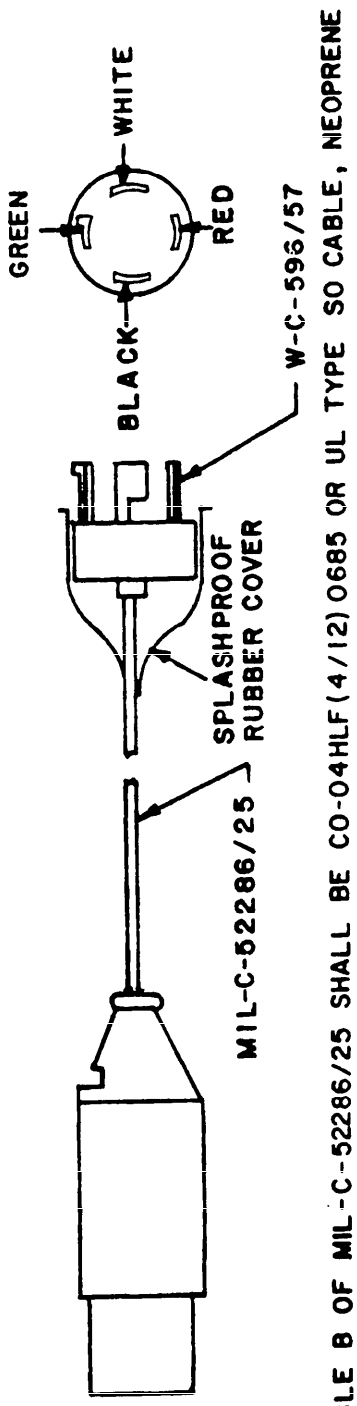
Custodians:
Army - ME
Air Force - 99

Review activity:
Army - MD

Preparing activity:
Army - ME

Project 4610-0088

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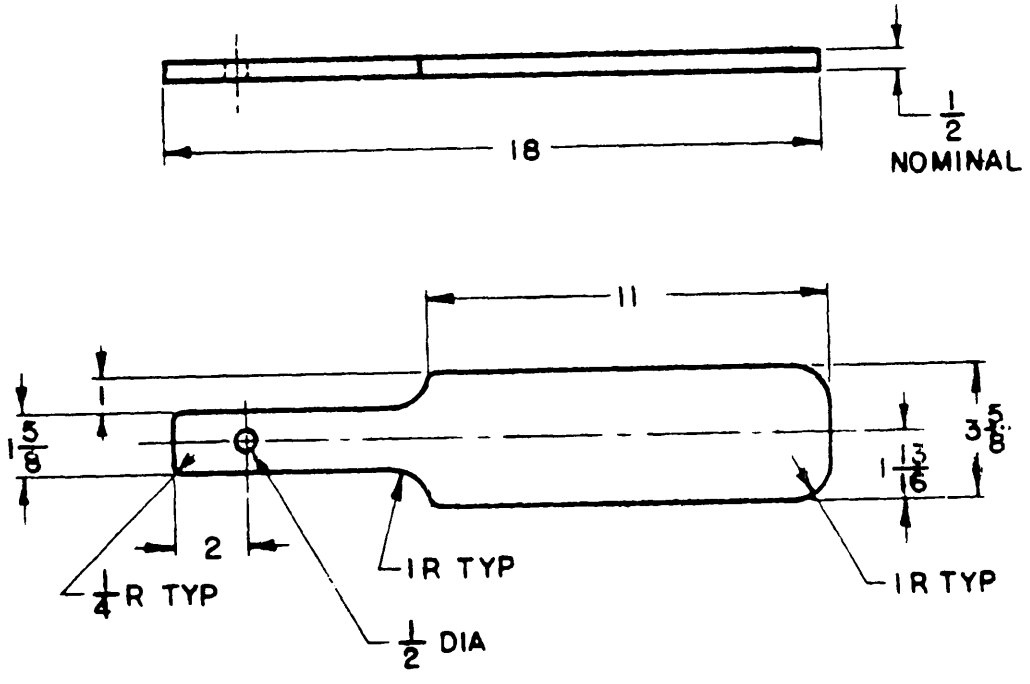


NOTE: CABLE B OF MIL-C-52286/25 SHALL BE CO-04HLF(4/12)0685 OR UL TYPE SO CABLE, NEOPRENE JACKET, 600 V, 60° C, TYPE SO-4, (4/12)(65 x 30).

FIGURE 1. Cable assembly, power, electrical, 4-conductor, one end w/watertight female plug connector, other end w/locking type male plug connector.

X-2117A

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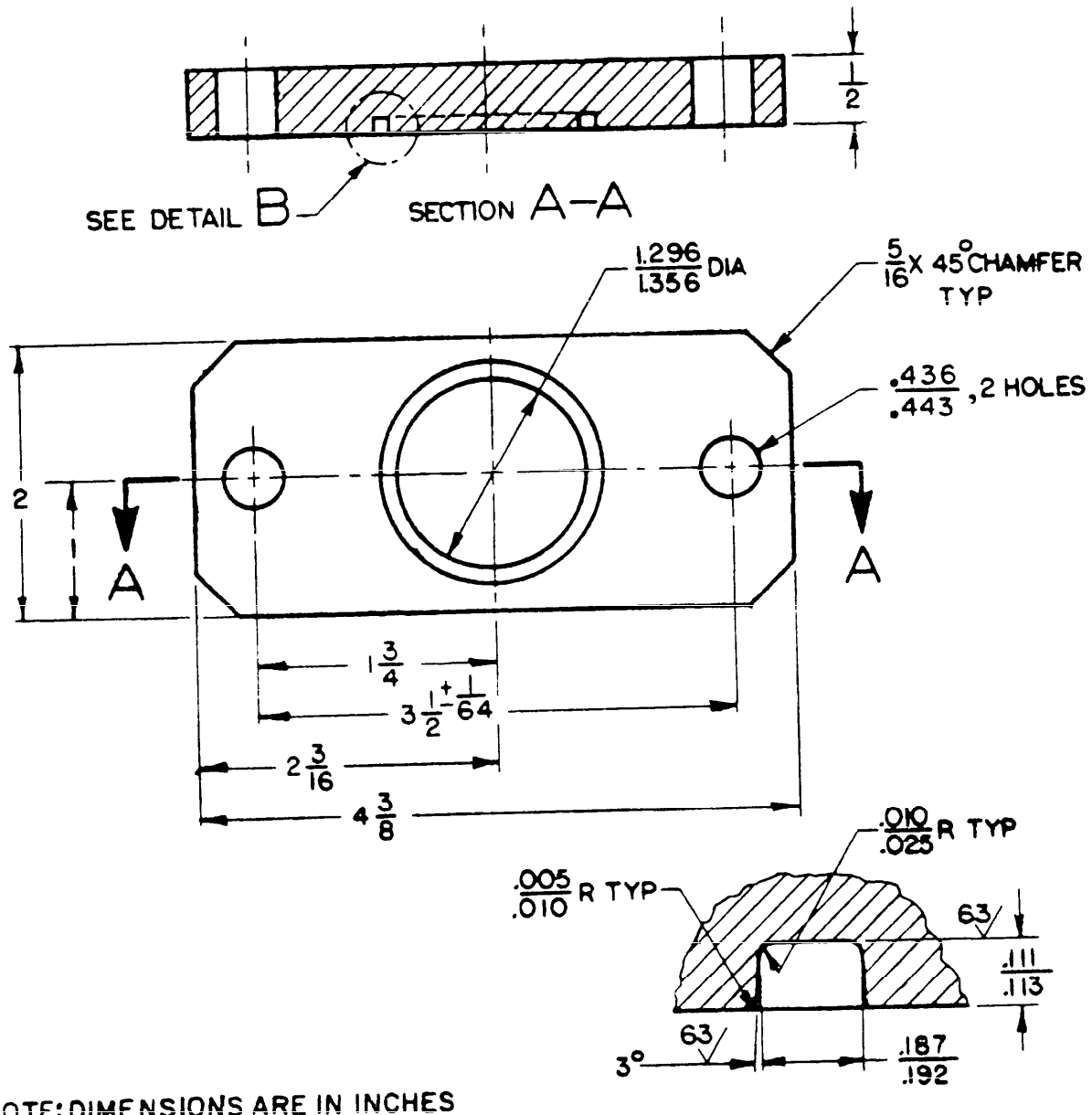
MATERIAL: REDWOOD

NOTE: DIMENSIONS ARE IN INCHES
ALL DIMENSIONAL TOLERANCES TO BE ± 1/8
UNLESS OTHERWISE SPECIFIED

FIG. 2. PADDLE

CX-1123

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NOTE: DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED TOLERANCES SHALL BE ± 1/32 INCH

DETAIL B

MATERIAL: ASTM-B209-6061-T6 ALUMINUM

FIG 3. FLANGE, BLIND

CX-1053

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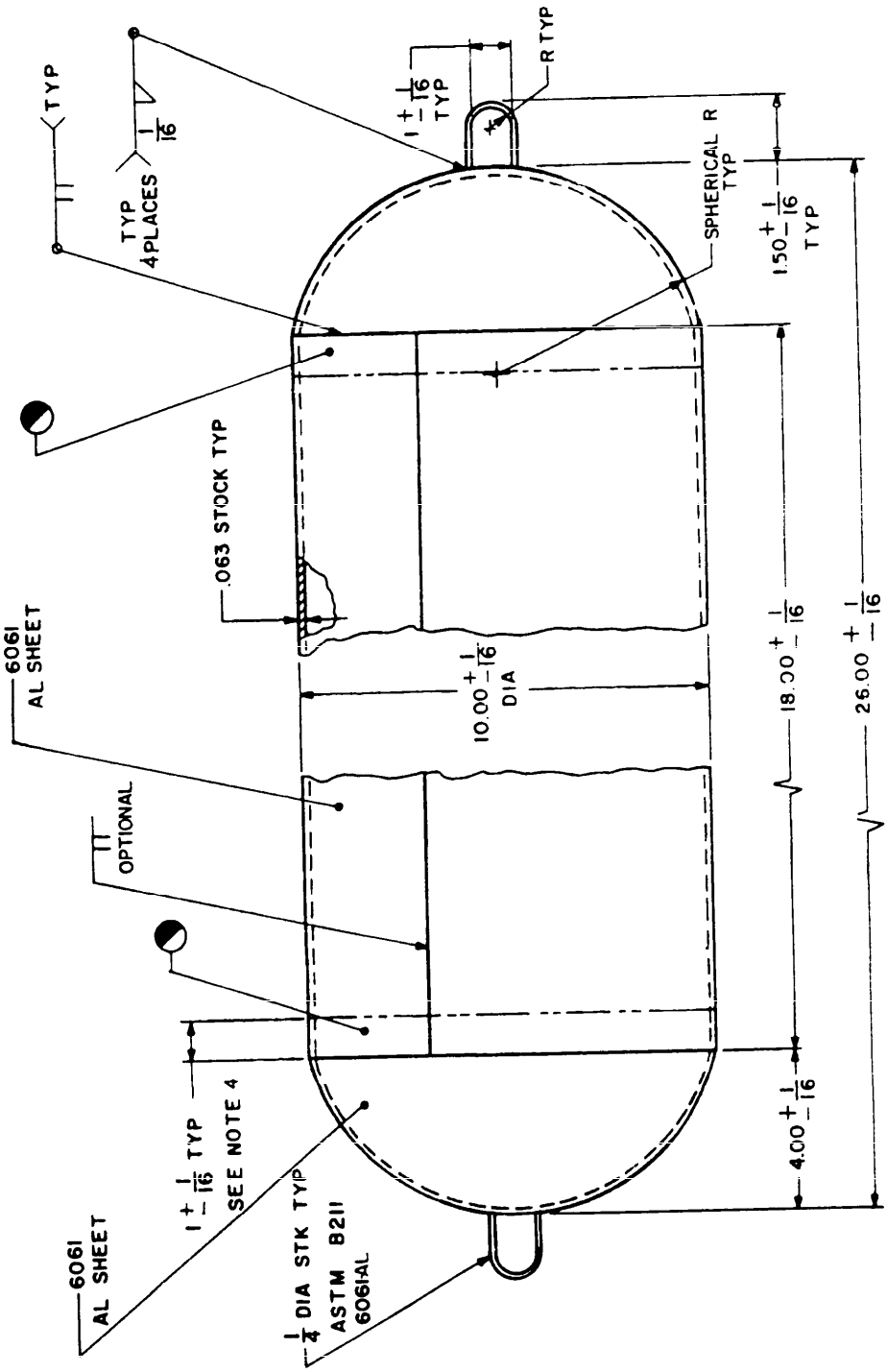
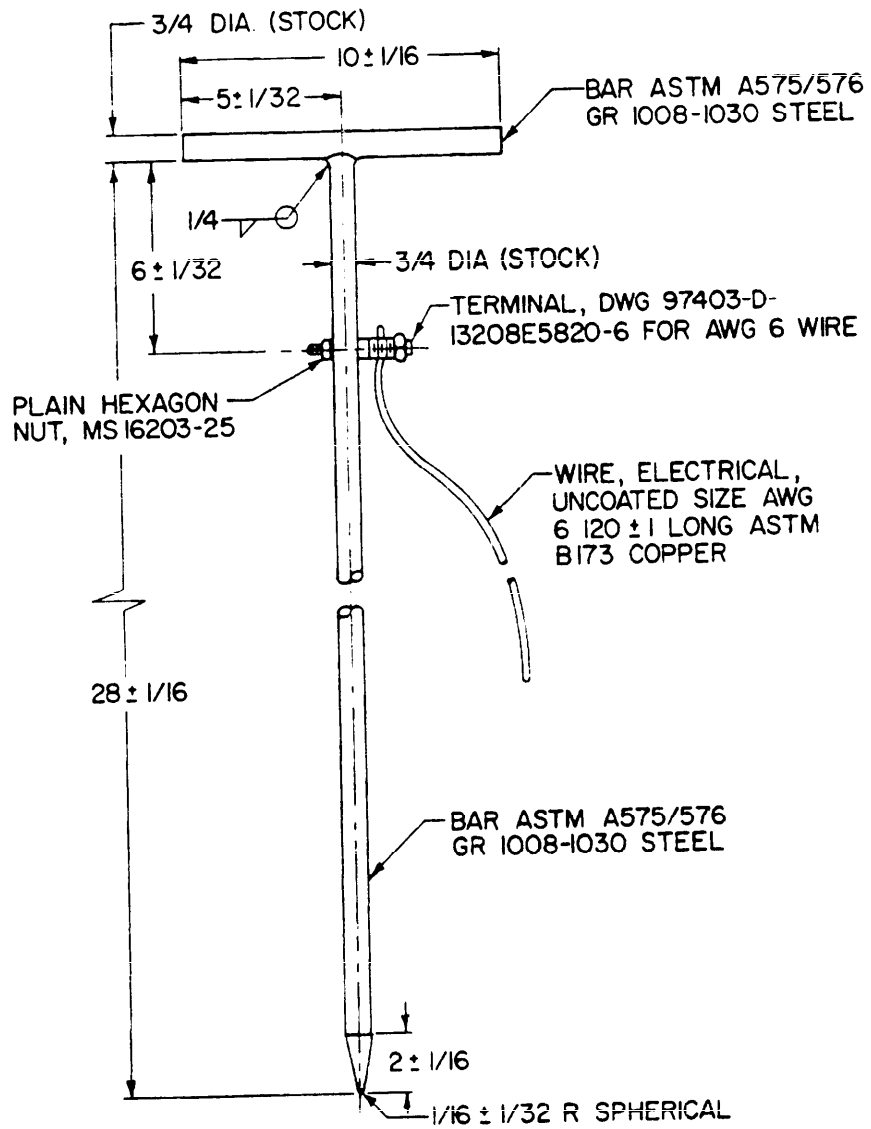


FIGURE 4 FLOAT X-291A

MATERIAL: ALUMINUM ALLOY ASTM B209, 6061 TEMPER OPTIONAL

1. DIMENSIONS ARE IN INCHES.
2. FLOAT TO BE WATERTIGHT
3. PAINT IN ACCORDANCE WITH MIL-T-704, TYPE B RED COLOR NO. 11105 PER FED-STD-595
4. AREAS MARKED ◔ PAINT IN ACCORDANCE WITH MIL-T-704, TYPE B WHITE COLOR NO. 17875 PER FED-STD-595

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NOTES:
1. DIMENSIONS ARE IN INCHES.
2. AFTER WELDING, HOT-DIP ZINC PLATE ALL PARTS, EXCEPT WIRE, TERMINAL, NUT, IN ACCORDANCE WITH ASTM-A123.

Figure 5. ELECTRICAL GROUND ROD.

DX290D

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-W-52500C Water Purification Unit, Electric-Motor-Driven, Erdlator-Type Clarifier, Diatomite-Type Filter, Van-Body-Mounted, 3,000 GPH

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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PENALTY FOR PRIVATE USE $300

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Development Command
ATTN: DRDME-DS
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